

Section 1: Number Theory

1. Circle the numbers in the table below that are divisible by 3.

699630	6661	5052
777	9991332	40000044
66666	783209	0

2. Circle the numbers in the table below that are divisible by 4.

7840	1000	5748324
6661	5423	80000052
5432	4325	0

3. Write a 10-digit number that is divisible by 12. The digits cannot all be the same number.

4. Why is it impossible to create a number that is divisible by 4 with only odd digits?

5. Order the digits 1, 9, and 2 to create a number that is divisible by 3, but not by 12.

Section 2: Real Algebra

1. Classify and solve

$$4x + 5 = 21.$$

2. Classify and solve

$$\frac{1}{x-2} + 7 = 10.$$

3. Classify and solve

$$x^2 - 7x - 78 = 0.$$

Section 3: Complex Arithmetic

1.

$$7 \cdot (7 + 6i) =$$

2.

$$(2 + 6i)^* =$$

3.

$$(2 - 8i) \cdot (8 + 7i) =$$

4.

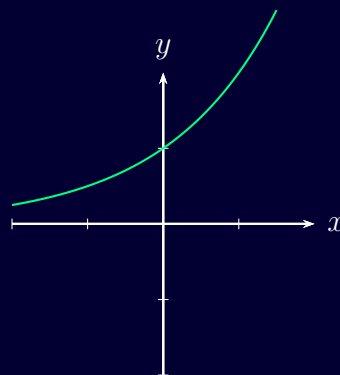
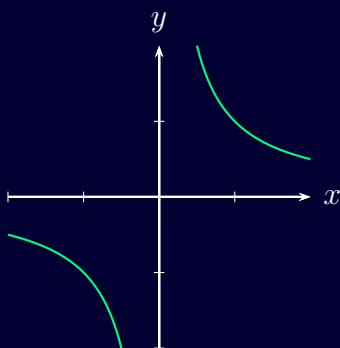
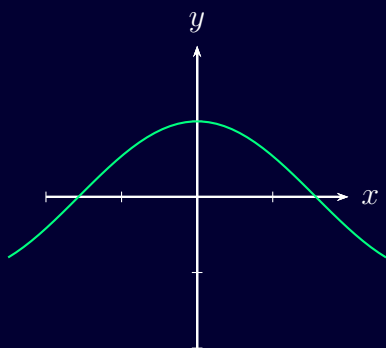
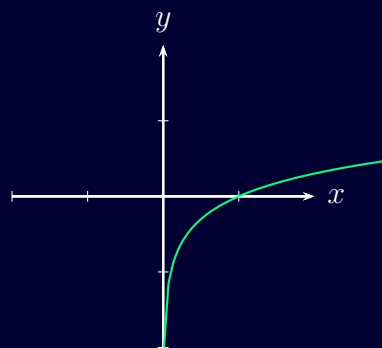
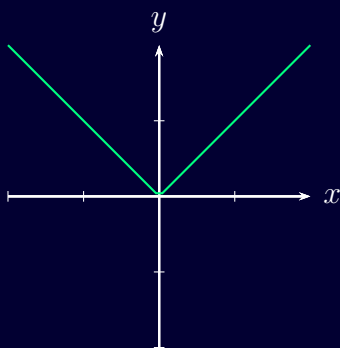
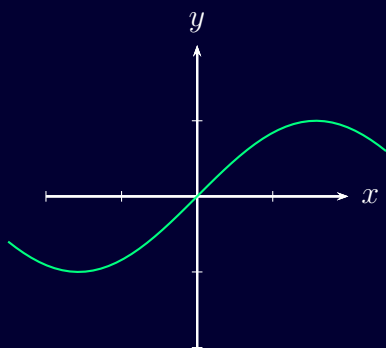
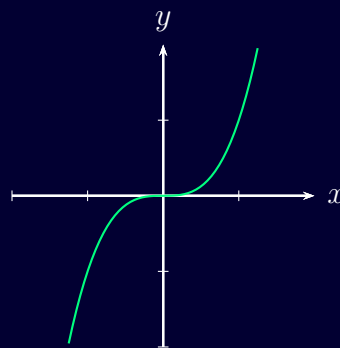
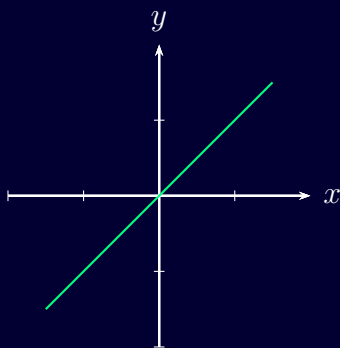
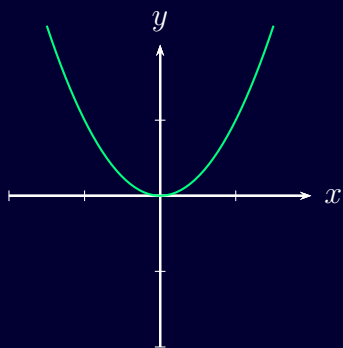
$$(9 + 2i) \cdot (9 - 2i) =$$

5.

$$\frac{2 + 3i}{1 - 2i} =$$

Section 4: Coordinate Geometry

1. Write the function that each graph corresponds to.



Section 5: Calculus

1. Find the derivative of

$$f(x) = \sin(\tan(\sec(\cos(x)))).$$

2. Find the derivative of

$$f(x) = x^{2022}.$$

3. Solve

$$\int \sec^2(x) \, dx.$$

Section 6: Challenge

1. Solve

$$\int 2xe^{x^2} dx.$$